# Sach Agarwal

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 8.89 |
| Aklank Public School, Kota | Class XII (CBSE) | 2024 | 92.40% |
| St. Mary's School , Najibabad | Class X (ICSE) | 2022 | 96.80% |

## WORK EXPERIENCE

**Summer Research Intern, IIT Bombay ||** LOR - Prof. Ganesh Ramakrishnan, IIT BOMBAY

- Co-authored a reader-friendly Machine Learning book covering **20+ core algorithms and concepts**, bridging foundations with practical applications for students, professionals and aspiring practitioners **through intuitive explanations and real-world examples.**
- Developed end-to-end content on **regression, classification and clustering**, incorporating scikit-learn implementations, mathematical intuition, algorithmic workflows and practical coding examples supported by visual illustrations and real-world case studies**.**
- Tailored the **book for an Indian audience** using Indian terminologies, examples, case studies and industry-specific applications.

**Founder's Office, PICAPOOL**

- Led grant applications and global fundraising outreach by engaging **1,000+ investors, accelerators, incubators and startup scouts** across the **US, Europe, UAE, Japan and Singapore**, securing selections in the **BeVisioneers Fellowship, Seaside Startup Summit (Armenia) and Trusity**, **resulting in 25+ lakh non-dilute grant funding** while operating bootstrapped.
- Developed investor-ready financial models, including a **5-year P&L, revenue forecasts, unit economics, fundraising deck and legal documentation** to support strategic fundraising, investor due diligence and long-term business planning.
- **Conducted** competitive benchmarking of **10+ hyperlocal networking platforms** by analyzing founding journeys, business models, growth trajectories, GTM strategies and competitive positioning to formulate Picapool's market entry and expansion strategy.

## PROJECTS

**RetainX: AI-Powered Retention & Digital Transformation Strategy for Quick Commerce | Self-Project**

- **Developed** an AI-powered digital transformation framework for India's **95,000 Cr** quick-commerce industry using **500K+ transactions**, **RFM/CLV segmentation**, **AS-IS/TO-BE redesign**, and AI initiatives projected to reduce churn by **10-15%**.
- Engineered a predictive churn model using **Python, SQL, XGBoost and feature engineering**, achieving **0.778 AUC** and **2.2× customer targeting lift** while designing interactive Power BI dashboards and executive KPI frameworks for data-driven decision making.
- Designed a comprehensive business case comprising **ROI, NPV, payback, risk assessment and GenAI adoption roadmap** projecting **121.7 Cr five-year NPV**, **2.0× ROI** and a **17-month payback period** through personalized customer retention strategies.

**Credit Card Customer Profitability Analytics | American Express Strategy Case Challenge**

- Developed a customer profitability framework by analyzing a **500,000+ record dataset** with **23 non-PII features**, deriving a profitability score using revenue from spend, revolving balances, expected credit losses, rewards and benefit costs.
- Built a predictive ranking model to identify the **top 20% most profitable customers**, achieving a **0.929 leaderboard score** through feature engineering, ensemble learning, model evaluation, hyperparameter optimization and business-driven feature selection.
- Translated insights into **targeting strategies** by prioritizing **high-value customer segments** through data-driven recommendations.

**AI Grant Application Automation Pipeline | PICAPOOL**

- Analyzed **50+ startup grant applications** to identify repetitive fundraising workflows, then engineered a **3-agent AI pipeline** that automated grant discovery, application scheduling, personalized follow-up generation, deadline tracking and CRM updates.
- Built an **AI Contact Discovery Agent** to scrape and consolidate information on **500+** grants, accelerators, incubators, and relevant contacts from public sources, generating a structured database with eligibility criteria, deadlines, and contact details.
- Developed an **AI Outreach & Scheduler Agent** that generated personalized follow-up emails, tracked applications, automated reminders and synchronized deadlines, reducing manual effort by **70%+** while supporting outreach across **100+ organizations.**

## SKILLS

**Technical Skills:** Python, SQL, C++, C, Machine Learning (Regression, Classification, Clustering, Ensemble Learning, Feature Engineering, Hyperparameter Tuning, Cross Validation, Model Evaluation), Generative AI Tools, Statistical Analysis, Digital Transformation

**Tools:** Microsoft Excel, PowerPoint, Word, Power BI, Tableau, Git, GitHub, Jira, Notion, Figma, Miro, R, Google Slides, Canva

## POSITIONS OF RESPONSIBILITY

**Head, Public Relations, Outreach and Marketing, Finance and Consulting Club**

- Led a team of **10 coordinator**s and **15+ volunteers** to organize institute-wide initiatives promoting awareness of careers in finance, consulting, and product management across IIT Hyderabad, **while managing the club's social media presence.**
- Conducted targeted outreach to **50+ consultants, venture capitalists, product managers and investment bankers** to organize speaker sessions, networking events, panel discussions, industry interactions across multiple domains at IIT Hyderabad.

**Core, Events and Public Relations & Outreach (PRO), Tinkerers' Lab**

- Led a team of **30+ volunteers** in planning, organizing and executing the **Under25 Summit**, featuring student performances and **a live musical performance**, while securing **Portronics** as the Official Gadget Sponsor for THRUST through strategic corporate outreach.
- Managed personalized email campaigns for **15,000+ students, professors and workshop participants** while ideating and creating social media content, including **TL Wrapped** and promotional initiatives to enhance outreach and engagement.
- Developed the **product design, content strategy and user experience** of the **Tinkerers' Lab website** by structuring content placement, optimizing information architecture, enhancing user navigation and improving content discoverability.

**Consulting Associate, Finance and Consulting Club || Head Internship Coordinator, Biotechnology and Bioinformatics ||**

**Production Coordinator, Inter IIT Sports Meet || Career Cell Coordinator, Office of Career Services**

## RELEVANT COURSES & CERTIFICATIONS

Introduction to Entrepreneurship, Introductory Economics, Data Structures & Applications, Introduction to Artificial Intelligence, Probability & Random Variables, Introduction to Programming, R & Python for Biologists, Biostatistics, Spoken Japanese Basics

**Certificates:** BCG Strategy Consulting Job Simulation and BCG Introduction to Strategy Consulting Job Simulation

## ACHIEVEMENTS & EXTRACURRICULARS

**Top 2% Performer among 17,500+ participants**, American Express Strategy Case Challenge 2026-27

**International Rank 9** - SOF International General Knowledge Olympiad (IGKO) and **3rd Place** - General Quiz (Diesta), IIT Hyderabad.

**1st Place** - Campus Case Challenge (FCC) and **75-Hour Short Film Making Competition** (MILAN), IIT Hyderabad.

**3rd Place, Water Rocketry Competition (MILAN)**, Inter-Hostel Championship, IIT Hyderabad